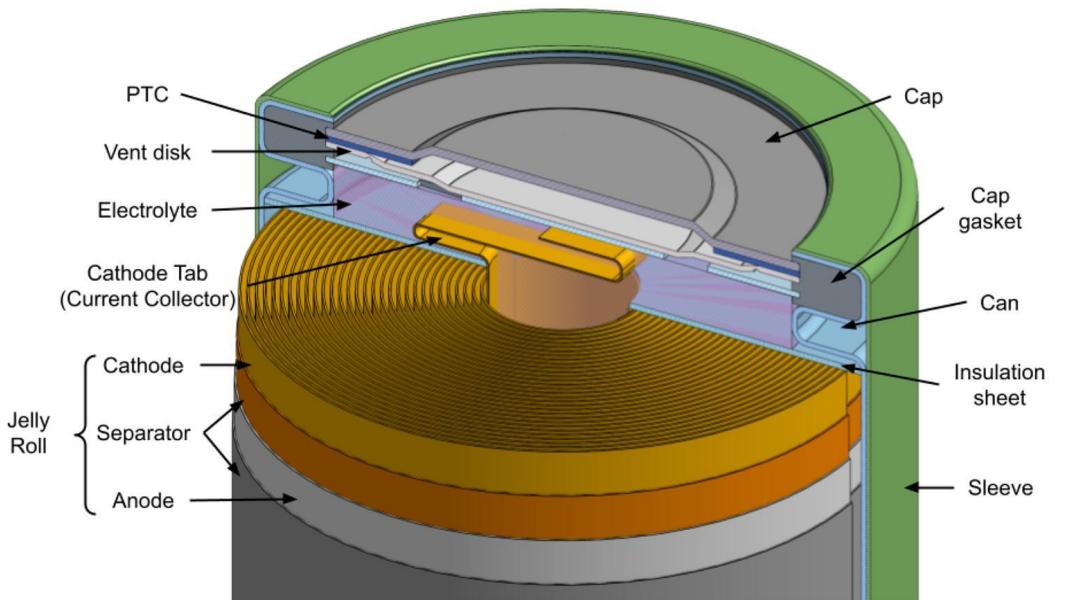


BROKEN QUALITY PRESS PRESENTS

Introduction to Lithium-Ion Cell Components & Defects

The whole is a function of its parts

An essay by Jim Lehner



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Introduction to Lithium-Ion Cell Components & Defects

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First edition

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Introduction to Lithium-Ion Cell Components & Defects

With the transition to electric aviation upon us, original equipment manufacturers (OEMs¹) in the aerospace industry face challenges that their ground-based electric vehicle (EV) counterparts did not. The increased complexity of flight comes with increased regulation from organizations like the Federal Aviation Administration (FAA). Certification through compliance with these agencies is simplified when internal and external manufacturing processes reliably produce products of world-class quality, *on-target with minimum variance*.² While this definition of quality applies equally to the airframe and avionics, it is of particular importance to the battery pack and the lithium-ion cells that compose it. Without the battery pack, electric aviation does not exist, and this emerging industry cannot take flight.

Composed of a network of individual cells arranged in series-parallel configurations, quality of the battery pack is bound to the quality of individual cells. Thus, in order to realize world-class quality at the pack level, OEMs must first realize world-class quality at the cell level. To achieve this, design, manufacturing, and quality teams must possess unified knowledge of the components that constitute individual cells and the defects that cause cell failure. Such knowledge establishes a foundation upon which cross-functional collaboration in product and process development is possible. It establishes a

¹ OEM refers to manufacturers of any battery-powered device, vehicle, or system.

² This definition of quality is from Dr. Genichi Taguchi in September 1968.

common language through which failure modes can be addressed before they become problems and ensures that safety risks present at the cell-level have no route for expression at the battery-pack level.

This document acts as an introduction to lithium-ion cells. Through an exploration of the primary cell components, their functions, and failure modes, it establishes a broad understanding of the technology and how its components work together. It is with this understanding that collective efforts to produce cells and battery packs of world-class quality become a reality of production rather than a goal that is constantly being chased.

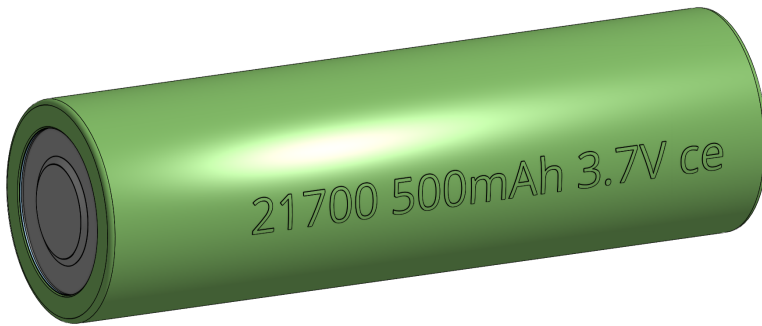


Figure 1. The geometry of the cylindrical lithium-ion cell is encoded in its naming convention. For the 21700 cell shown here, the outside diameter of the cell is 21 mm and the height is 70.0 mm.

The components of lithium-ion cells

Before discussion of cell and pack quality can begin, it is useful to establish an understanding of the cell components, their functions, and their relationships. At a high-level, cylindrical lithium-ion cells are composed of seven components: the can, the cap, the cathode, the anode, the separator, the electrolyte, and the current collectors (cathode tab and anode tab). While other components contribute to cell performance and safety, such as the vent disk,

positive temperature coefficient (PTC) device, and cap gasket, the preceding list constitutes a starting point for understanding the components of an individual cell and how it works.

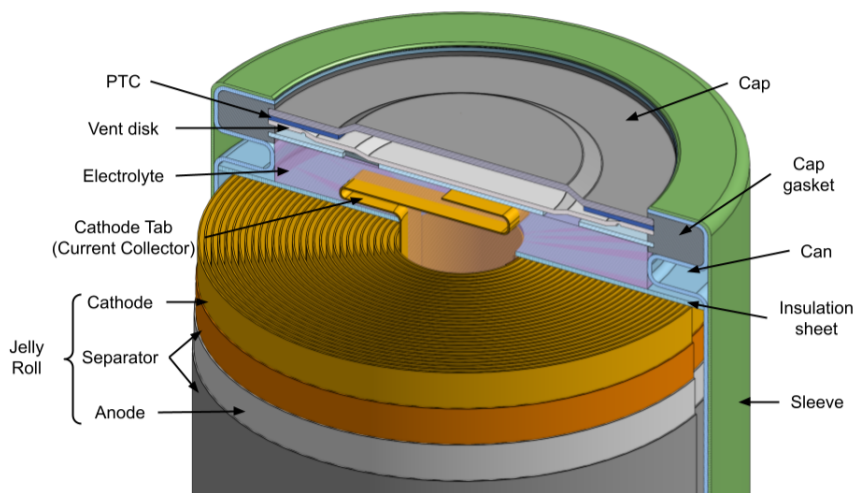


Figure 2. Sectional view of a lithium-ion cell reveals that there is more here than meets the eye.

The outer housing of a lithium-ion cell is called the can. Typically made of 300 series stainless steel, the can defines the characteristic shape of an individual cell. Although the can's primary function is to contain cell components within a functional package, it also serves as the bottom electrical terminal.

The can is typically wrapped with a sleeve made of insulating material, such as polyvinyl chloride (PVC) or polyethylene terephthalate (PET). Regardless of the material, the sleeve helps prevent short circuits. More often than not, it is printed with important battery information, such as type, capacity, voltage, and safety warnings. In addition to containing the cell components and serving as an electrical terminal, the can also shields the internal components from environmental stresses such as impacts and vibrations.

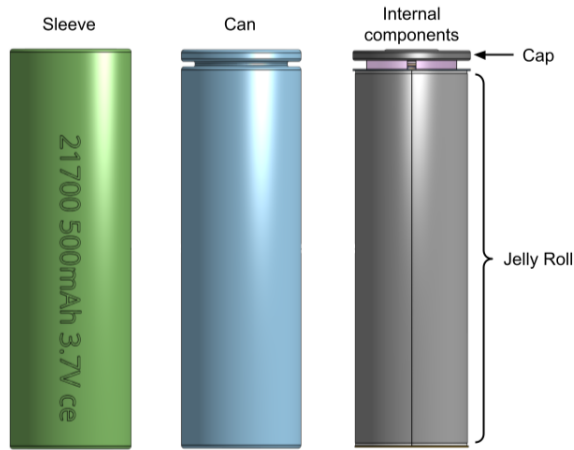


Figure 3. The sleeve and can define the characteristic shape of the cell. Together they contain and protect the cells internal components during transport and operation.

An often-overlooked but vital component of the cell is the cap. Composed of a series of subcomponents held in place by a nonconductive plastic insert or gasket, the cap is designed to satisfy physical, electrical, and safety requirements. The most obvious function of the cap is to seal the cell, isolating it from the surrounding environment. It also plays a functional role in facilitating electrical connections.

Viewed from the outside, what is typically thought of as the cap is actually the cell's positive terminal. This is the point through which between the cell and the external circuit. Typically unseen but not to be forgotten are the cap's safety features. These features, including the positive temperature coefficient (PTC) device, are designed to protect the cell under abnormal operating conditions and reduce the risk of overheating, thermal runaway, and explosion.³

³ P. G. Balakrishnan et al., "Safety Mechanisms in Lithium-Ion Batteries," *Journal of Power Sources* 155 (2006): 401–414.

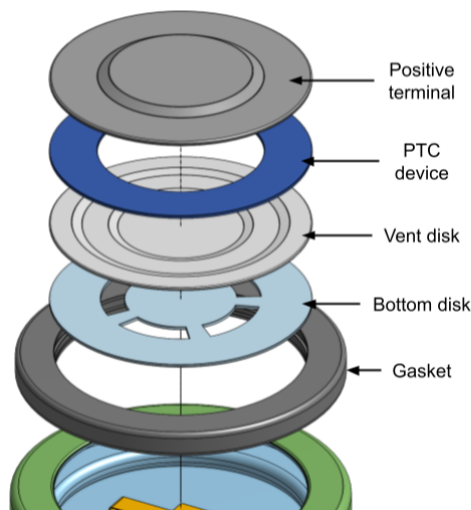


Figure 4. The cap is composed of a series of sub-components that aim to satisfy physical, electrical, and safety requirements.

Consuming the majority of a cell's internal space are the components of the jelly roll. The jelly roll is composed of the cathode, the anode, and the separator. The cathode is the positive electrode in the jelly roll. It is made from a lithium metal oxide material that is coated onto a foil substrate. The anode is the negative electrode in the jelly roll and is typically made of graphite.⁴

The cathode and the anode are isolated from each other by the separator. Typically composed of a microporous polymer such as polyethylene (PE) or polypropylene (PP), the separator prevents physical contact between the cathode and the anode. Although the microporous structure of the separator blocks the flow of electrons between the cathode and anode, it allows lithium ions to pass through. This movement of lithium ions is essential to the cell's function. It enables the electrochemical reactions that allow the cell to store and release electrical energy.

⁴ Guangxu Zhang et al., "Internal Short Circuit Mechanisms, Experimental Approaches and Detection Methods of Lithium-Ion Batteries for Electric Vehicles: A Review," *Renewable and Sustainable Energy Reviews* 141 (May 2021).

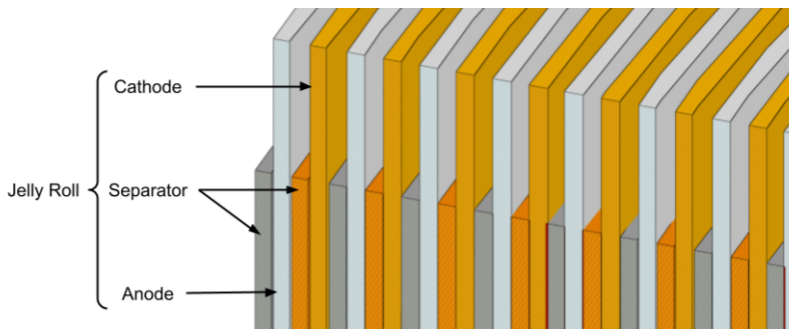


Figure 5. Constituting the majority of the internal volume of a cell is the jelly roll. The jelly roll is composed of a tightly wound bundle of alternating cathode, separator, and anode layers.

After inserting the internal components of a cell into the can but before it is sealed, the cell is filled with electrolyte. The primary function of the electrolyte is to facilitate the movement of lithium ions during charging and discharging. The electrolyte is typically composed of a lithium salt dissolved in an organic solvent. The ratio of the solvent dictates the flammability and auto-ignition temperature of the electrolyte.⁵

Each cell contains two current collectors, the cathode tab and the anode tab. These tabs connect the electrodes to the cell's external terminals, providing a pathway for electrons to flow between the electrodes and the external circuit. When the cell is discharging, lithium ions flow from the anode (negative electrode) through the electrolyte to the cathode (positive electrode). When the cell is charging, lithium ions flow through the electrolyte from the cathode to the anode.⁶ While the cathode tab is typically made from aluminum, the anode tab is typically made from copper. Aluminum is well suited to the cathode because of its electrochemical stability at the higher potentials encountered during cell operation, while copper is well suited to the anode

⁵ Sahithi Maddipatia et al., "Safety Analysis of Lithium-Ion Cylindrical Batteries Using Design and Process Failure Mode and Effects Analysis," University of Maryland.

⁶ Xiang-Yan Yao et al., "Tab Design and Failures in Cylindrical Li-Ion Batteries."

because of its electrochemical stability at lower potentials.⁷

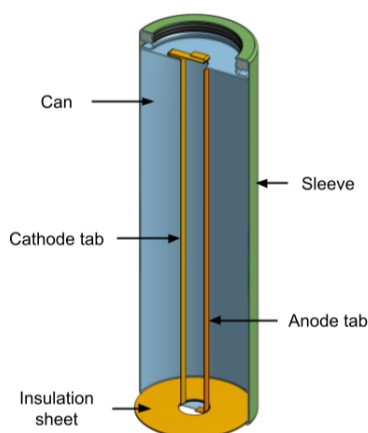


Figure 6. The cathode and anode tabs connect the cathode and anode to the external circuit.

The cell defect landscape

With the prior discussion of the components of cells complete, the next step is to establish an understanding of the cell failure landscape. Broadly speaking, cell failures occur across a continuum. At the extreme ends of the continuum are performance degradation and safety events. These are inversely related to one another with regard to likelihood and severity. Where performance degradation has a high likelihood but low severity, safety events have low likelihood but high severity. The continuum is bifurcated by functional failures. These failures occur with moderate likelihood and moderate severity. However, some functional cell failures can manifest in ways that, if left unchecked, can lead to safety events.

⁷ M. Bonnemann et al., “Batteries Safety: Recent Progress and Current Challenges,” *Frontiers in Energy Research* 7 (2019): 71.

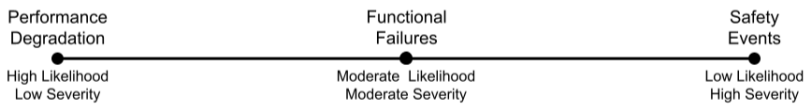


Figure 7. The cell failure continuum is bounded on the left by performance degradation (high likelihood but low severity) and on the right by safety events (low likelihood but high severity).

The least severe and most likely to occur of the three failures modes is performance degradation. It is an unavoidable result of the electrochemical reactions that occur during normal cell operation, gradually reducing the cell's ability to store and deliver electrical energy.⁸ Consumers and industry alike have come to see performance degradation as an unavoidable, albeit inconvenient, consequence of using batteries as an energy source.

The performance degradation of individual cells is reflected in the performance of the battery pack. While this manifest in a variety of ways, one such example of its impact is on the Battery Management System (BMS). The BMS plays an essential role in maintaining battery pack balance. As individual cells age, differences in their electrical and thermal behavior can increase across the pack. This places greater demands on the BMS as it works to manage differences in cell capacity, voltage, and state of charge. Left unchecked, these differences can accelerate degradation and shorten the useful life of the battery pack.

Occurring at moderate frequencies and severity are functional failures. These are often the result of physical defects or contamination that occurs during manufacturing. Functional failures can be further classified in accordance with two sub-categories: open-circuit failures and short-circuit failures.

Open-circuit failures result from open-circuit defects. These defects are caused by a discontinuity or break in the electrical pathway of a cell. Like any functional failure, the defects that cause open-circuit failures at the cell level can propagate into failures at the battery-pack level.

⁸ Peter Attia, "Perspective: Battery Quality at Scale," Glimpse.

Short-circuit failures occur when unintended electrical pathways form between the cathode and anode of a cell, allowing current to bypass the intended external circuit. Knowledge of short-circuit defects is of particular importance because of their impact on battery-pack performance. In extreme cases, a short-circuit defect can cause cells connected in series with the defective cell to become overcharged as the battery system attempts to maintain the required voltage.⁹ Similarly, a short-circuit defect can cause current to flow from cells connected in parallel with the defective cell, causing those cells to discharge as the battery system attempts to maintain voltage balance.¹⁰ Moreover, short-induced thermal runaway can lead to catastrophic cell failure and, by extension, battery-pack failure.¹¹

Open-circuit defects	Short-circuit defects
Tab tear	Metallic particle contamination
Terminal corrosion	Insufficient separator overhand
Premature activation of PTC	Electrode wrinkle
Tab weld failure	Electrode buckle

Figure 8. Functional failures occur due to open-circuit defects and short-circuit defects. These defects typically occur during manufacturing, although the resulting failure can be latent.

Regardless of the functional subclassification, both open-circuit and short-circuit defects at the cell level can manifest as failures at the pack level. Two such examples are tab weld failures, an open-circuit defect, and electrode buckling, a short-circuit defect.

Tab welds connect the internal current collectors of a cell to its external terminals. This provides the electrical pathway through which current flows into and out of the cell. When a tab weld failure occurs, whether in production

⁹ “Parallel and Series Connections,” *Encyclopedia of Electrochemical Power Sources*.

¹⁰ M. Dubarry et al., “Cell-Balancing Currents in Parallel Strings of a Battery System.”

¹¹ L. Lu et al., “A Review on the Key Issues for Lithium-Ion Battery Management in Electric Vehicles.”

or due to excessive mechanical or thermal stress, it manifests as an open-circuit failure. The consequences of such a failure become immediately obvious.

Battery packs consist of many cells arranged in series-parallel configurations. Any cell in a series connection that exhibits an open-circuit failure will cause all cells in that series to become inoperable, producing the “Christmas tree” effect. While tab weld failures do not necessarily pose an existential threat to the battery pack, they can prevent the affected series string from delivering current. This can reduce the pack’s available voltage and power output. Moreover, depending on the configuration, it may cause the pack to become completely inoperable.

Another example of functional failure is electrode buckling. Electrode buckling refers to the mechanical deformation of the electrode layers inside a cell. This can be caused by mechanical stresses that develop during manufacturing or as a consequence of thermal expansion and contraction. Regardless of the cause, electrode buckling affects the performance and safety of individual cells. In its most severe form, it can cause internal short circuits between the positive and negative electrodes of the jelly roll. These short circuits can generate localized heating that can lead to thermal runaway within the cell, potentially posing a threat at the pack level.

Bounding the severe end of the failure continuum are cell-level safety events. While the frequency of these events is low, their occurrence can have dramatic and irreversible consequences for both individual cells and battery packs. All efforts to eliminate the occurrence of safety events should be considered a top priority for cross-functional teams.

Safety events most notably result from internal short circuits. These unwanted and unintended connections between the positive and negative electrodes within a cell can be caused by manufacturing defects or external events, such as crashes. In their most extreme form, internal short circuits can lead to thermal runaway within a cell. Thermal runaway is a self-sustained chain reaction of heat generation within a cell. It causes a rapid increase in temperature and pressure that can result in fire or explosion. Thermal runaway of a single cell can quickly spread to surrounding cells. This can lead

to the complete loss of the battery pack or, even worse, the loss of the vehicle that the pack is powering.

Regardless of the domain and the specific defects that cause cell failure, the prior discussion highlights the interconnected nature of cell and battery pack quality. While the failure of individual cells, when not integrated into a battery pack, can prove catastrophic, the same failures, when those cells are integrated into a battery pack, can prove existential. This is why manufacturing and quality teams responsible for manufacturing battery packs require practical knowledge of how individual cells work and fail. By establishing a foundational understanding of cell performance, teams can develop effective incoming inspection methods to screen for potential failure modes. Moreover, if battery pack level failures do occur, teams with this foundational knowledge will be better equipped to identify their root causes.

Summary

As a *network of interdependent components working together to accomplish an aim*,¹² the quality of the eCTOL or eVTOL battery pack is a product of its requisite parts. Of those parts, the most prominent and important are the lithium-ion cells. A working knowledge of cells, including their fundamental components, their functions, and the defects that cause failures, is essential to manufacturing battery packs of world-class quality.

While such a statement may read trite, world-class quality is of particular importance to the fledgling electric aviation industry. With more rigorous oversight than its ground based counterparts, electric flight faces many challenges. Those challenges, whether related to certification or manufacturing at scale, are eased when design, manufacturing, and quality teams speak the same language. That language is established through a foundational understanding of individual cells, their functions, and how they fail.

¹² W. Edwards Deming, *The New Economics for Industry, Government, Education*, (Cambridge, MA: MIT Press, 1993).

